Handling missing data

MCAR (Missing Completely At Random) means data is missing with no relation to any variable — purely random.
Example: a survey form lost due to printing error.
It’s the best-case scenario for missing data because it doesn’t bias results.

MAR (Missing At Random) means data is missing for a reason related to observed variables, not the missing one itself.
Example: Older people skip an income question more often — missingness depends on age, which we know.
It’s manageable using statistical methods like imputation or regression.

MNAR (Missing Not At Random) means data is missing because of the value that’s missing itself.
Example: People with very high income don’t report their income — missingness depends on the hidden value.
It’s the hardest type to handle since you can’t fix it without making assumptions or collecting more data.

SMOTE :
SMOTE creates synthetic (new) examples of the minority class rather than simply duplicating existing ones.

⚖️ Why SMOTE is Needed

In classification problems, an imbalanced dataset looks like this:

Class	Count
0(majority)	900
1(minority)	100

- If you train a model on such data, it might just predict all zeros — because that gives it 90% accuracy!
  SMOTE helps fix this by creating more samples for class 1, so the model learns both classes properly.

scikit learn

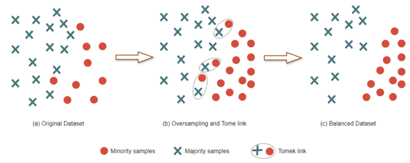
🔬 How SMOTE Works (Step-by-Step)
1. Pick a sample from the minority class.
2. Find its k nearest neighbors (usually k=5) in the feature space.
3. Randomly select one of these neighbors.
4. Generate a new synthetic sample between the two points
   xnew=xi+rand(0,1)×(x neighbor−xi)
5. Repeat until the desired balance is reached.

So SMOTE interpolates between real samples to create synthetic but realistic new samples.

- Imbalanced datasets impact the performance of the machine learning models and the Synthetic Minority Over-sampling Technique (SMOTE) addresses the class imbalance problem by generating synthetic samples for the minority class.

Data Imbalance in Classification Problem

- Data imbalance in classification refers to skewed class distribution, hindering machine learning models' performance.
- Techniques like Oversampling, Undersampling, Threshold moving, and SMOTE help address this issue.
- Handling imbalanced datasets is crucial to prevent biased model outputs, especially in multi-classification problems.



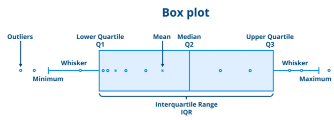
HOW SMOTE WORKS

BOX - PLOT and OUTLIERS

What is box plot

- A Box Plot is a data visualization that summarizes a dataset’s distribution.
- It’s an efficient way to spot patterns and identify outliers i.e data points that stand out from the rest due to their extreme values.

Key Components of a Box Plot



- Minimum: Smallest value in the dataset excluding outliers.
- First Quartile (Q1): Middle value of the lower half of the data (25th percentile).
- Median (Q2): Middle value that splits the dataset into two equal parts (50th percentile).
- Third Quartile (Q3): Middle value of the upper half of the data (75th percentile).
- Maximum: Largest value in the dataset excluding outliers.

Understanding the Box Plot

A Box Plot consists of three main parts:

- Box: The box spans from the first quartile (Q1) to the third quartile (Q3) representing the interquartile range (IQR).
  This box contains the middle 50% of the data with a line inside it representing the median (Q2).
- Whiskers: The lines extending from the box called whiskers shows the range of the data. They extend from Q1 to the minimum value and from Q3 to the maximum value excluding outliers.
- Outliers: Data points that fall outside the whiskers are considered outliers. These are marked separately and represent values that are significantly higher or lower than the rest of the data.

How to Calculate Quartiles and Identify Outliers?

- Order the Data: Arrange the data from smallest to largest.
- Find the Median (Q2): The median divides the dataset into two halves:

  If the dataset has an odd number of data points, exclude the median from both halves.
  If the dataset has an even number of data points, the median is the average of the two middle values.
- Calculate the Quartiles:

  First Quartile (Q1): Median of the lower half of the data.
  Third Quartile (Q3): Median of the upper half of the data.
- Identify Outliers:

  Interquartile Range (IQR) is calculated as: IQR = Q3 - Q1
  An outlier is any data point that lies outside the bounds defined by:

  Lower Bound: Q1 - 1.5 * IQR
  Upper Bound: Q3 + 1.5 * IQR
- Any data point beyond these bounds is considered an outlier.